

Claudius Heyer

Curriculum Vitae

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Education

- July 2024 – **Postdoc**, *Universität Paderborn*
Oct. 2019 – **Postdoc**, *Universität Münster*
June 2024
Mar. 2015 – **PhD in Mathematics**, *Humboldt-Universität zu Berlin*, Title: *Applications of parabolic Hecke algebras: parabolic induction and Hecke polynomials*
28.06.2019
Oct. 2012 – **M. Sc. in Mathematics**, *Humboldt-Universität zu Berlin*
Jan. 2015
Oct. 2009 – **B. Sc. in Mathematics**, *Humboldt-Universität zu Berlin*
Aug. 2012

Positions, Fellowships

- July 2024 – **Postdoc**, *Universität Paderborn*
Oct. 2022 – **Postdoc**, *Universität Münster*, no teaching load
June 2024
Oct. 2019 – **Research Assistant**, *Universität Münster*
Sep. 2022
Feb. 2018 – **Berlin Mathematical School, Phase II (PhD student)**
Sep. 2019
Apr. 2016 – **Research Assistant**, *Humboldt-Universität zu Berlin*
Sep. 2019

Papers, Preprints

- 2025 **Second Adjointness**, *Preprint*, arXiv:2508.14598v1, joint with Manuel Hoff, Sarah Meier and Michael Spieß
Appendix to: M. Hoff, S. D. Meier, M. Spieß, *On the Right Derived Functors of Ordinary Parts*
2024 **6-Functor Formalisms and Smooth Representations**, *Preprint, submitted*, arXiv:2410.13038, joint with Lucas Mann
2023 **The Geometrical Lemma for Smooth Representations in Natural Characteristic**, *Preprint, submitted*, arXiv:2303.14721

- 2022 ***The Left Adjoint of Derived Parabolic Induction***, *Math. Z.* **305** 46(2023), doi:10.1007/s00209-023-03385-5
- 2021 ***On the Smooth Part Functor***, *Preprint*, arXiv:2108.05262
- 2021 ***On the Decomposition of Hecke Polynomials over Parabolic Hecke Algebras***, *J. Theor. Nomb. Bordx.* **34** (2022) no. 3, 941–997, doi:10.5802/jtnb.1235
- 2021 ***Localization of the Parabolic Hecke Algebra at a Strictly Positive Element***, *Preprint*, arXiv:2103.16949
- 2021 ***Parabolic Induction via the Parabolic pro- p Iwahori–Hecke Algebra***, *Represent. Theory* **25** (2021), 807–843, doi:10.1090/ert/585
- 2019 ***Applications of parabolic Hecke algebras: parabolic induction and Hecke polynomials***, *Dissertation, Humboldt-Universität zu Berlin, Mathematisch-Naturwissenschaftliche Fakultät*, 2019, doi:10.18452/20137

Research Talks

- 25.09.2026 **A 6-Functor Formalism for Smooth mod p Representations**, *Wuppertal*, Germany, HEGAL 2026
- 15.07.2026 **Second Adjointness for Smooth mod p Representations**, *Seoul*, South Korea, KIAS virtual seminar (Online via Zoom)
- 20.03.2026 **On Second Adjointness for mod p Representations**, *Essen*, Germany, $\ell = p$: New developments in p -adic representation theory
- 18.06.2025 **A 6-Functor Formalism for Smooth Representations**, *Lille*, France, Workshop on Iwasawa Theory and Representations of p -Adic Groups
- 16.01.2025 **A 6-Functor Formalism for Smooth mod p Representations**, *Essen*, Germany, Oberseminar
- 18.12.2024 **A 6-Functor Formalism for Smooth mod p Representations**, *Wuppertal*, Germany, Seminar Algebra und Topologie
- 13.12.2024 **A 6-Functor Formalism for Smooth mod p Representations**, *Bielefeld*, Germany, BIREP-Seminar
- 08.10.2024 **A 6-Functor Formalism for Smooth mod p Representations**, *Tokyo*, Japan, Workshop on Shimura varieties, representation theory and related topics, 2024
- 29.05.2024 **On Smooth mod p Representations of p -adic Reductive Groups**, *Paderborn*, Germany, Research Seminar
- 22.02.2024 **The Geometrical Lemma in Natural Characteristic**, *Lille*, France, Séminaire “Arithmétique”

- 24.10.2023 **Six Functors and Smooth mod p Representations**, *Münster*, Germany, Mittagsseminar
(Online via Zoom)
- 06.06.2023 **The Geometrical Lemma in Natural Characteristic**, *Wuppertal*, Germany, Oberseminar
- 25.05.2023 **The Geometrical Lemma in Natural Characteristic**, *Essen*, Germany, Oberseminar
- 20.01.2022 **The Left Adjoint of Derived Parabolic Induction**, *San Diego, California*, USA, UCSD Number Theory Seminar
(Online via Zoom)
- 24.03.2021 **Derived Products and the Left Adjoint of Derived Parabolic Induction**, *Rennes*, France, Colloque Tournant 2021 du GDR Théorie de Lie Algébrique et Géométrie
(Online via Zoom)
- 12.01.2021 **Derived Products and the Left Adjoint of Derived Parabolic Induction**, *Münster*, Germany, Mittagsseminar
- 26.08.2019 **Parabolic Induction for pro- p Iwahori–Hecke Modules**, *Marseille*, France, Conference: Buildings and Affine Grassmannians
- 28.05.2019 **Decomposition of Hecke Polynomials**, *Münster*, Germany, Mittagsseminar

--- Conferences, Summer Schools, Workshops

- Sep. 22–25 2026 **Workshop: Mod p HEcke algebras and GALois moduli (HEGAL)**, *University of Wuppertal*, Wuppertal, Germany
- Mar. 18–20 2026 **Workshop: $\ell = p$: New developments in p -adic representation theory**, *University of Duisburg–Essen*, Essen, Germany
- Feb. 9–13 2026 **p -adic aspects of the Langlands program**, *CIRM*, Marseille, France
- Sep. 1–3 2025 **Semiorthogonal Decompositions for Representations of Algebraic Groups**, *Bielefeld University*, Bielefeld, Germany
- Aug. 27–29 2025 **Early Number Theory Researchers (ENTR) Workshop 2025**, *University of Paderborn*, Paderborn, Germany
- Aug. 4–8 2025 **BIREP Summer School on Finite Tensor Categories 2025**, *Hotel im Schlosspark*, Willebadessen, Germany
- June 18–20 2025 **Workshop: Iwasawa Theory and Representations of p -Adic Groups**, *University of Lille*, Lille, France
- Oct. 7–11 2024 **Workshop on Shimura varieties, representation theory and related topics 2024**, *University of Tokyo*, Tokyo, Japan
- Sep. 9–13 2024 **Conference: Integral Structures in Geometry and Representation Theory**, *University of Paderborn*, Paderborn, Germany

- June 24–28 2024 **Workshop: Geometrization of the Archimedean Local Langlands Correspondence**, *Maison du Kleebach*, Munster, France
- June 10–14 2024 **Masterclass: Continuous K-theory, dualizable and rigid categories**, *University of Copenhagen*, Copenhagen, Denmark
- Mar. 25–27 2024 **Mathematics Münster Midterm Conference**, *University of Münster*, Münster, Germany
- Jul. 31 – Aug. 4 2023 **Summer School: The Automorphic Side of the p -adic Langlands Program**, *University of Wuppertal*, Wuppertal, Germany
- Feb. 13–17 2023 **Conference: Number Theory Meets p -Adic Representations**, *University of Münster*, Münster, Germany
- July 18–22 2022 **Focused Research Event: Smooth representations of $GL(n, \mathbb{Q}_p)$ in natural characteristic**, *University of Oxford*, Oxford, United Kingdom
- Mar. 16–18 2022 **Workshop: Higher Algebra**, *TU Darmstadt*, Darmstadt, Germany
- Sep. 27 – Oct. 1 2021 **Conference: Representation Theory's Hidden Motives**, *University of Münster*, Münster, Germany
- Apr. 26–30 2021 **Spring School towards a mod p Langlands correspondence**, *Essen*, (Online via Zoom)
- Mar. 24–26 2021 **Colloque Tournant 2021 du GDR Théorie de Lie Algébrique et Géométrie**, *University of Rennes 1*, (Online via Zoom)
- Aug. 26 – Sep. 6 2019 **Conference: Buildings and Affine Grassmannians**, *CIRM, Marseille, France*
- June 3–7 2019 **Conference: Representation Theory and D-Modules**, *University of Rennes 1, Rennes, France*
- Oct. 8–11 2018 **CENTRAL Workshop: Automorphic Techniques in Arithmetic Geometry**, *Humboldt-Universität zu Berlin, Berlin, Germany*
- Sep. 2–8 2018 **Autumn School: Arithmetic of Differential Equations**, *Łukęcin, Poland*

Teaching experience

- SS 2026 **Teaching Assistant**, *University of Paderborn*, Analysis for Computer Scientists, Lecturer: Dr. Job J. Kuit
- WS 2025/26 **Lecturer**, *University of Paderborn*, Linear Algebra for Computer Scientists
- SS 2025 **Organizer**, *University of Paderborn*, Seminar: Introduction into the Theory of Quiver Representations
- WS 2024/25 **Lecturer**, *University of Paderborn*, Triangulated and Derived Categories
- WS 2023/24 **Organizer**, *University of Münster*, informal Reading Seminar: cuspidal representations of $GL_2(F)$ for $F/\mathbb{Q}_p$ finite, Co-Organizer: Prof. Dr. Eugen Hellmann

- SS 2023 **Lecturer**, *University of Münster*, Smooth Representations of p -Adic Groups
- WS 2022/23 **Organizer**, *University of Münster*, Mittagsseminar
- SS 2022 **Teaching Assistant**, *University of Münster*, Introduction to Algebra, Lecturer: Prof. Dr. Eva Viehmann
- WS 2021/22 **Teaching Assistant**, *University of Münster*, Reflection Groups and Platonic Solids, Lecturer: apl. Prof. Dr. Lutz Hille
- SS 2021 **Organizer**, *University of Münster*, Oberseminar: Classification of Smooth mod p Representations
- SS 2021 **Teaching Assistant**, *University of Münster*, Galois Representations and (φ, Γ) -Modules, Lecturer: Prof. Dr. Peter Schneider
- WS 2020/21 **Teaching Assistant**, *University of Münster*, Local Galois Representations, Lecturer: Prof. Dr. Peter Schneider
- SS 2020 **Organizer**, *University of Münster*, Seminar: Finite Hecke Algebras, Co-Organizer: Prof. Dr. Peter Schneider
- SS 2020 **Organizer**, *University of Münster*, Oberseminar: Towards Eigenvarieties
- WS 2019/20 **Assistant**, *University of Münster*, Seminar: Mathematical Learning and Reflection, Main Instructor: PD Dr. Christian Serpé
- SS 2019 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Algebra/Number Theory and its Didactics, Lecturer: Prof. Dr. Elmar Große-Klönne
- WS 2018/19 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Algebra and Complex Analysis, Lecturer: Prof. Dr. Elmar Große-Klönne
- SS 2018 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Linear Algebra and Analytic Geometry II*, Lecturer: Prof. Dr. Elmar Große-Klönne
- WS 2017/18 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Linear Algebra and Analytic Geometry I*, Lecturer: Prof. Dr. Elmar Große-Klönne
- SS 2017 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Algebra and Number Theory, Lecturer: Dr. Daniel Skodlerack
- WS 2016/17 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Linear Algebra and Analytic Geometry I, Lecturer: Dr. Claudia Schillings
- SS 2016 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Algebra and Number Theory, Lecturers: Dr. Werner Kleinert and Dr. Frank Gounelas
- WS 2015/16 **Teaching Assistant**, *Humboldt-Universität zu Berlin*, Linear Algebra I (for Computer Scientists), Lecturer: Dr. Bernhard Gerlach

Organization

- Aug. 27–29 **ENTR Workshop 2025**, *University of Paderborn*, Co-organizers: Abhijit
2025 A. J., Dr. Lucas Mann, Dr. Ingmar Metzler, Tomáš Perutka, Luca Speciale

Nov. 2025 – **Paderborner Mathezirkel**, *University of Paderborn*, Organizer: Dr. Kerstin Hesse

Languages

German native

English fluent

French basic

Claudius Heyer
Paderborn, September 22, 2026